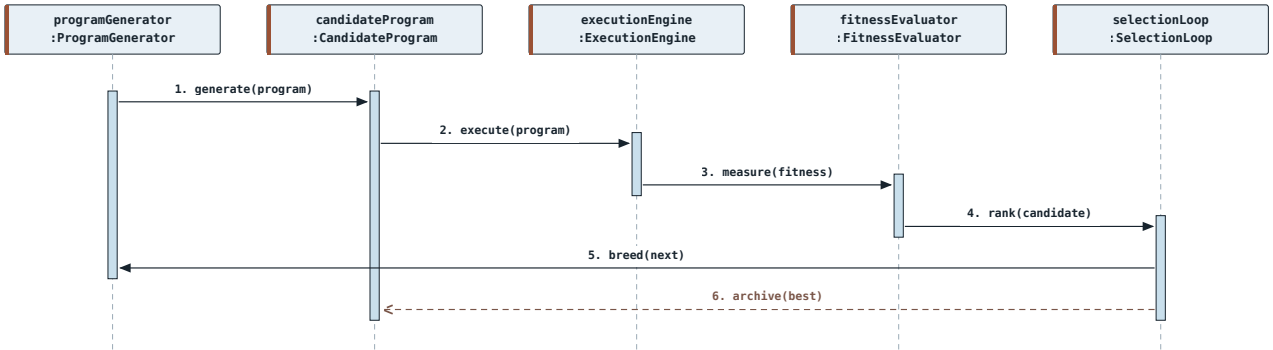
E07 Evaluator-guided program search

Evolution / OBJECT INTERACTION / ANALYTICAL COLLABORATION

Evolve executable candidates.

OBJECT INTERACTION

solid arrow: call/message · dashed open arrow: return/failure · analytical order, not an observed trace



POINTCUT / PATTERN INTERFACE

CONTEXT	Harness or harness-adjacent behavior is represented as executable programs.
TRIGGER	Candidate programs are generated, mutated, or selected.
STRUCTURE	Generator, executor, fitness evaluator, archive, and next-generation policy.
EFFECT	A next generation is selected from measured executable candidates.

CONTRACT / REQUIRED BEHAVIOR

REQUIRES	Candidate runtime and evaluation suite are isolated and available.
ENSURES	Every selected candidate has execution and fitness provenance.
INVARIANT	Evaluation budget and security boundaries hold across all candidates.
VIOLATION	Crash, timeout, unsafe code, or invalid fitness removes or quarantines candidate.

Sources: protocol/pattern_diagram_models.yaml (participants + messages); protocol/harness_pattern_specification.yaml (pointcut + contract).

Analytical collaboration only; no repository execution or message order is asserted.